GLOBAL FOOD SECURITY AND SUPPLY CHAIN ANALYSIS

1. Project Overview and Objective

To analyze global food trade data, this project cleans and combines four Cleaned Excel tables (shipments, products, ports and alerts) in Power BI. Delays at ports and customs can spoil food and cause losses, but the data is scattered and messy, so problem areas are hard to spot. The dashboards will show trade value, delays, inspection results and spoilage risk, helping to find bottlenecks and reduce food waste.

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2. Data Sources

- **Source Description and Timeline:** UN FAOSTAT Data Hub and the United Nations Comtrade Plus Database **and** 2025 & 2026.
- **Domain**: AGRI-FOOD Logistics.

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3. Problem Statement

To Clean and analyze global food trade performance using Excel and builds interactive Power BI dashboards that reveal trade value, tariff patterns, logistics efficiency, spoilage risk and port delays

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4. Attribute (Column /Features) Details:

TABLE -1 : FOOD_TRADE_TRANSACTIONS.

Attribute_Name	Data_Type	Description
Transaction_ID	String (Text)	Unique identifier for each shipment, e.g. TXN_100000
Product_ID	String (Text)	Code of the food product shipped
Log_Date	Date	Date the shipment was logged
Export_Country	Categorical	Country the food is shipped from
Import_Country	Categorical	Country the food is shipped to
Shipment_Weight_Tons	Numeric (Decimal)	Weight of the shipment in tons
Unit_Price_USD	Numeric (Decimal)	Price per unit in US dollars
Customs_Tariff_Pct	Numeric (Decimal)	Customs tariff charged on the shipment, in percent

Logistics_Mode	Categorical	Transport method (Ocean Freight, Air Cargo, Rail)
Port_ID	String (Text)	Code of the port used; links to Port_Directory
Crop_Season	Categorical	Season of the crop (Spring, Summer, Autumn, Winter)
Spoilage_Risk_Score	Numeric (Decimal)	Chance of the food spoiling, scored from 0 to 1
Inspection_Status	Categorical	Inspection result (passed, held, quarantined)
Carbon_Footprint_Score	Numeric (Decimal)	Environmental impact score of the shipment

TABLE -2: PRODUCT_CATALOG.

Attribute_Name	Data_Type	Description
Product_ID	String (Text)	Unique identifier for each product, e.g. PROD_701
Item_Name	String (Text)	Name of the food item, e.g. Basmati Rice
Food_Category	Categorical	Type of food (Dairy, Grains, Fruits)
Storage_Temp_Celsius	Numeric(Integer)	Required storage temperature in °C
Shelf_Life_Days	Numeric(Integer)	Number of days the product stays fresh

TABLE- 3: PORT_DIRECTORY.

Attribute_Name	Data_Type	Description
Port_ID	String (Text)	Unique identifier for each port, e.g. PRT_301
Port_Name	String (Text)	Name of the port
Country	Categorical	Country where the port is located
Max_Capacity_TEU	Numeric (Integer)	Maximum cargo capacity, in twenty-foot containers (TEU)
Operational_Status	Categorical	Current port status (Active, Under Maintenance)
Latitude	Numeric(Decimal)	North-south position of the port on the map
Longitude	Numeric(Decimal)	East-west position of the port on the map

TABLE -4: LOGISTICS_ALERTS.

Attribute_Name	Data_Type	Description
Transaction_ID	String (Text)	Shipment affected by the alert
Alert_Type	Categorical	Cause of the delay (Customs Hold, Labor Strike, Monsoon Delay, Port Congestion)
Delay_Hours	Numeric(Decimal)	Number of hours the shipment was delayed

TABLE-5 : CALENDER.

Attribute_Name	Data_Type	Description
Date	Date	Date of the Year
Year	Numeric (Whole Number)	Year of the Date
Month_No	Numeric (Whole Number)	Month Number of the Year
Month	String (Text)	Month Name of the Year
Quarter	Numeric(Whole Number)	Quarter of the Year

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5. Tools & Technologies

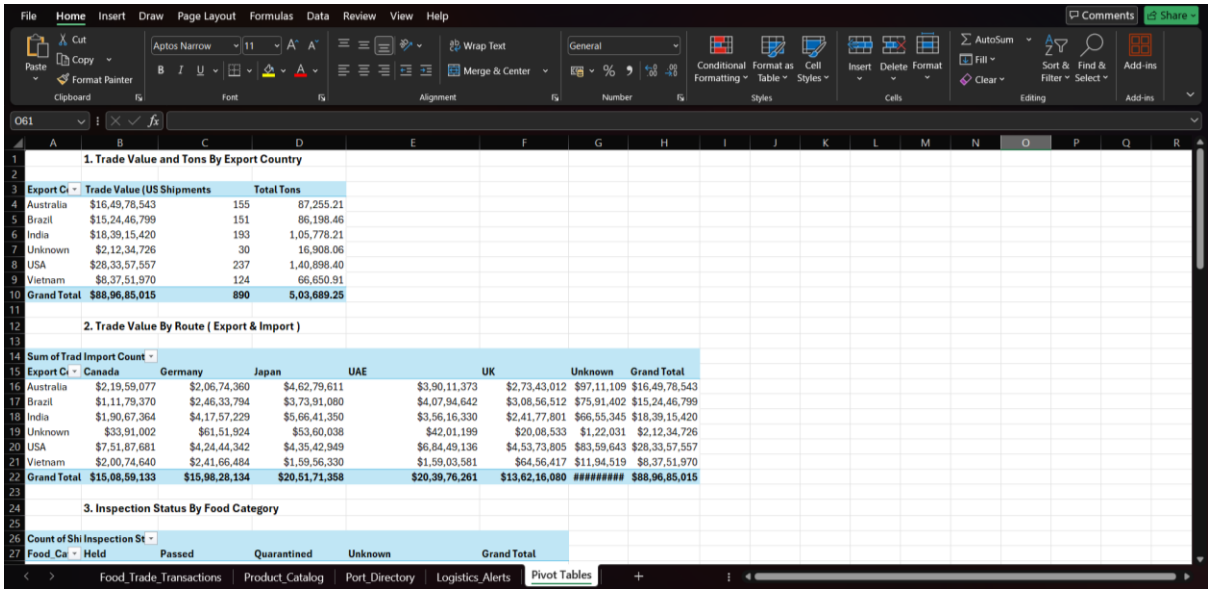
- **Excel :** Data cleaning, transformation, and Pivot Tables.
- **Power BI :** Data modeling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.



1. Trade Value and Tons By Export Country

Export Country	Trade Value (US$)	Shipments	Total Tons
Australia	$16,49,78,543	155	87,255.21
Brazil	$15,24,46,799	151	86,198.46
India	$18,39,15,420	193	1,05,778.21
Unknown	$2,12,34,726	30	16,908.06
USA	$28,33,57,557	237	1,40,898.40
Vietnam	$8,37,51,970	124	66,650.91
Grand Total	**$88,96,85,015**	**890**	**5,03,689.25**

2. Trade Value By Route (Export & Import)

Sum of Trade Value (US$) / Import Country	Canada	Germany	Japan	UAE	UK	Unknown	Grand Total
Australia	$2,19,59,077	$2,06,74,360	$4,62,79,611	$3,90,11,373	$2,73,43,012	$97,11,109	$16,49,78,543
Brazil	$1,11,79,370	$2,46,33,794	$3,73,91,080	$4,07,94,642	$3,08,56,512	$75,91,402	$15,24,46,799
India	$1,90,67,364	$4,17,57,229	$5,66,41,350	$3,56,16,330	$2,41,77,801	$66,55,345	$18,39,15,420
Unknown	$33,91,002	$61,51,924	$53,60,038	$42,01,199	$20,08,533	$1,22,031	$2,12,34,726
USA	$7,51,87,681	$4,24,44,342	$4,35,42,949	$6,84,49,136	$4,53,73,805	$83,59,643	$28,33,57,557
Vietnam	$2,00,74,640	$2,41,66,484	$1,59,56,330	$1,59,03,581	$64,56,417	$11,94,519	$8,37,51,970
Grand Total	**$15,08,59,133**	**$15,98,28,134**	**$20,51,71,358**	**$20,39,76,261**	**$13,62,16,080**	#########	**$88,96,85,015**

3. Inspection Status By Food Category

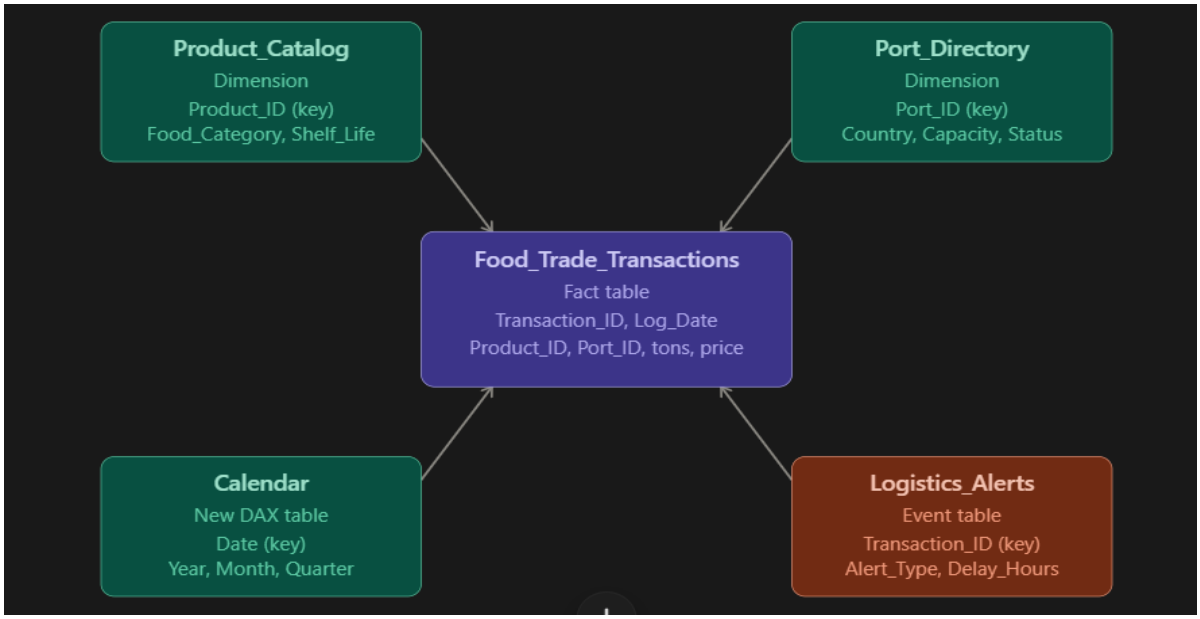
Count of Shipment ID / Inspection Status	Held	Passed	Quarantined	Unknown	Grand Total

Food_Trade_Transactions | Product_Catalog | Port_Directory | Logistics_Alerts | Pivot Tables

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FACT AND DIMENSION TABLE .

STAR SCHEMA :

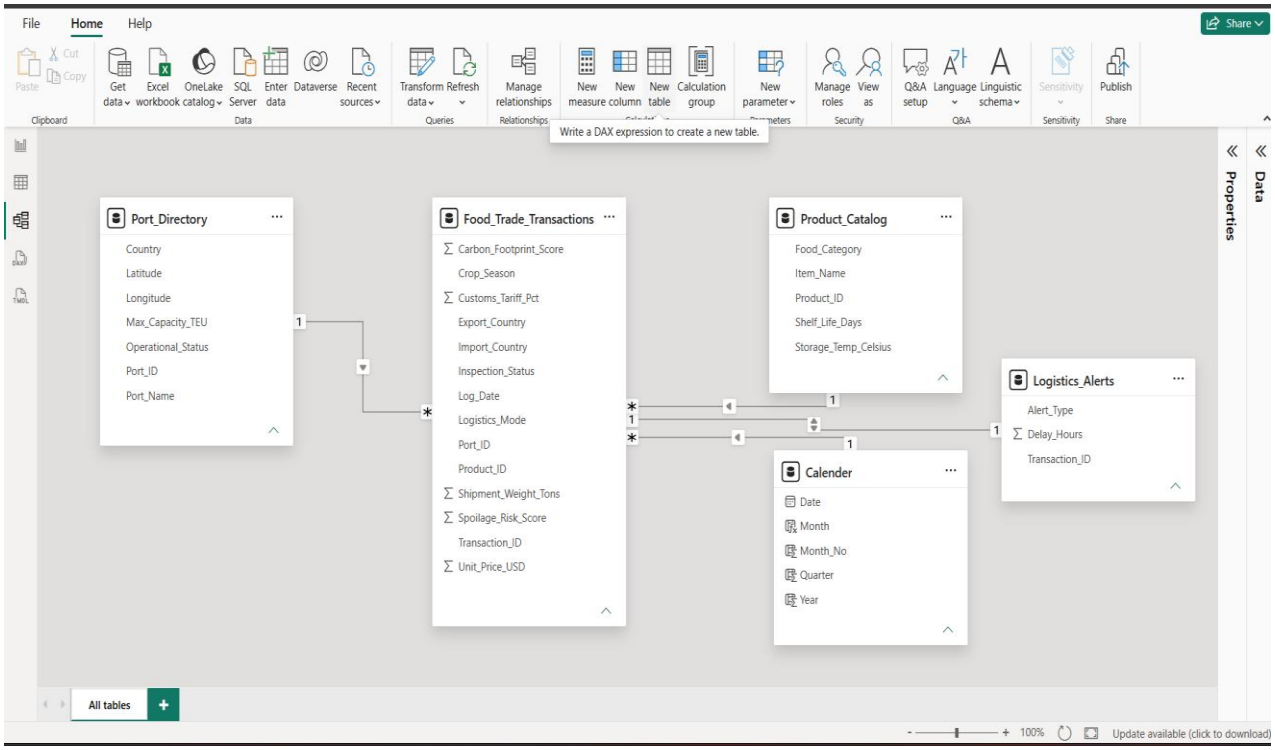


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7. Data Modelling and DAX (Power BI)

Data Model: Established relationships between tables , cardinality and Cross Filter Directions are Defined

SCREENSHOT OF DATA MODELING.



Food_Trade_Transactions (Port_ID)----Port_Directory (Port_ID)

Cardinality----Many to One Cross Filter Direction----Single

Food_Trade_Transactions (Product_ID)----Product_Catalog (Product_ID)

Cardinality----Many to One Cross Filter Direction----Single

Food_Trade_Transactions (Transaction_ID)----Logictics_Alerts (Transaction_ID)

Cardinality----One to One Cross Filter Direction----Both

Food_Trade_Transactions (Log_Date)----Calender (Date)

Cardinality----Many to One Cross Filter Direction----Single

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8. Analysis and Visualizations (Power BI)

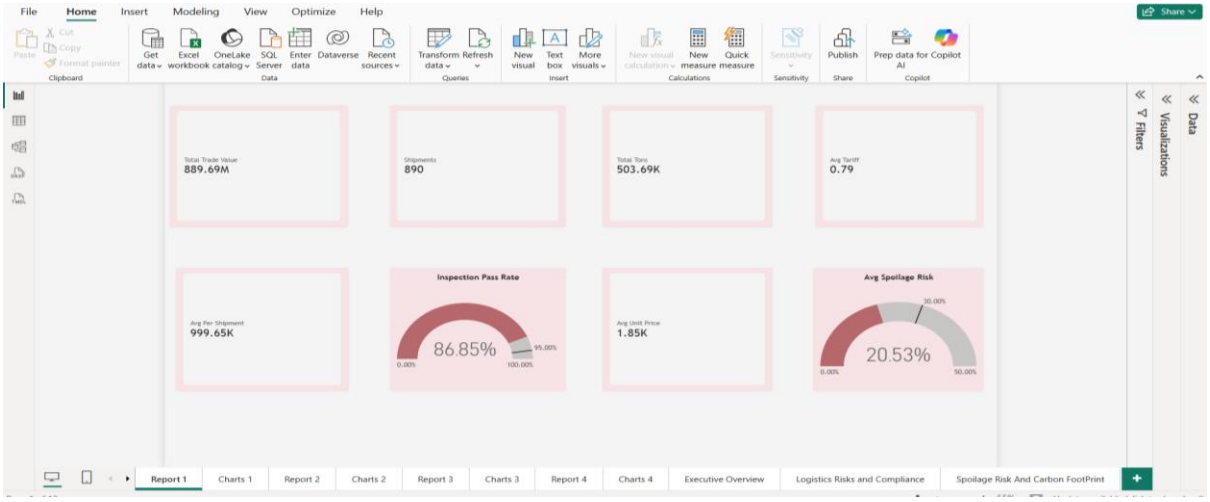
Dashboard Features:

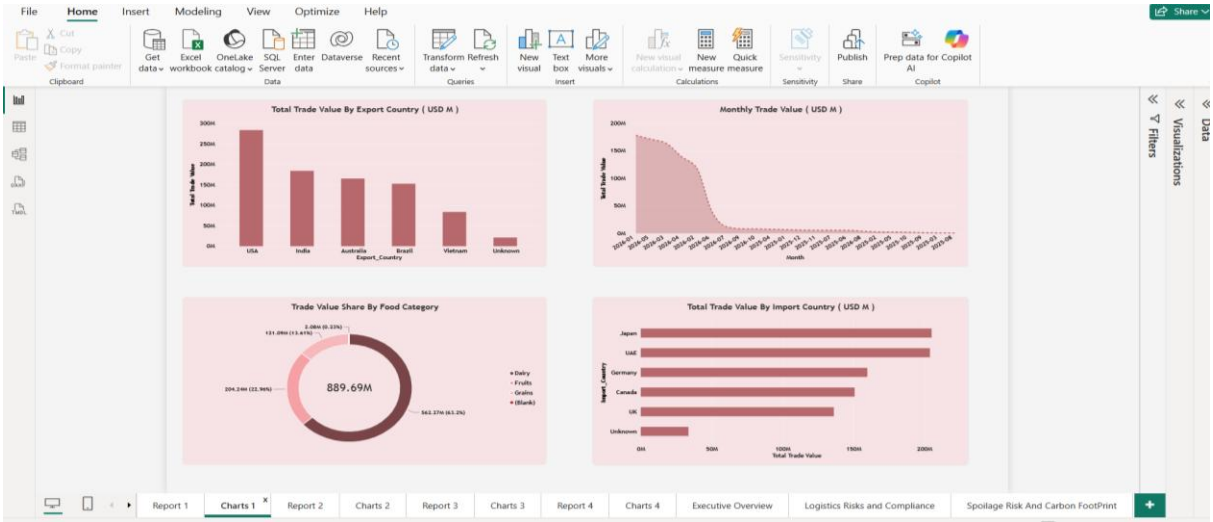
Many Cards and Gauges , Slicers ,Button Slicers are Used. Bookmarks are Created.

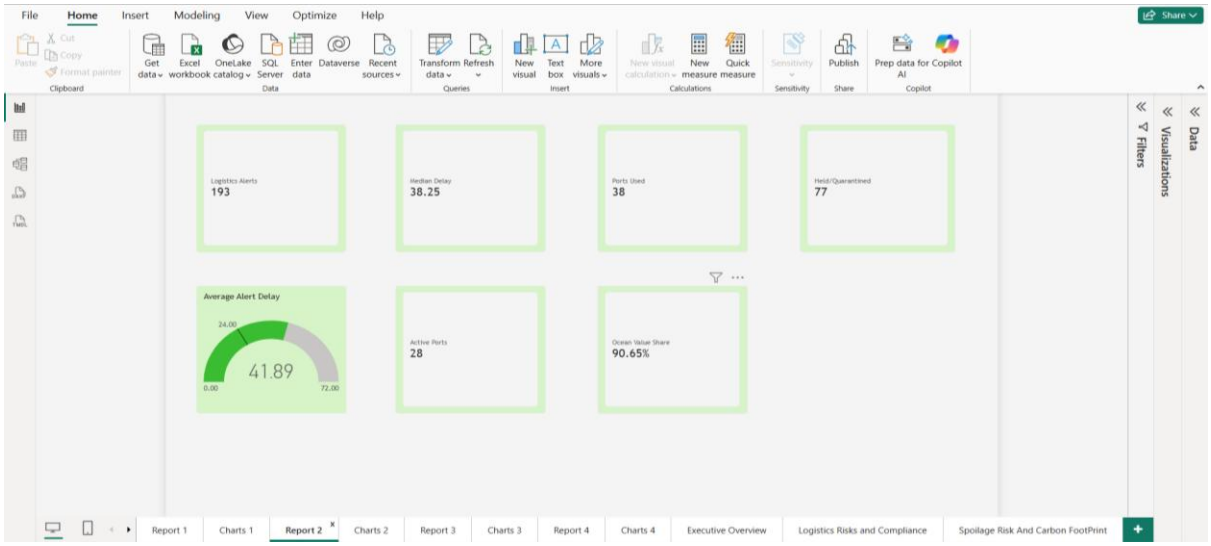
Charts Used :

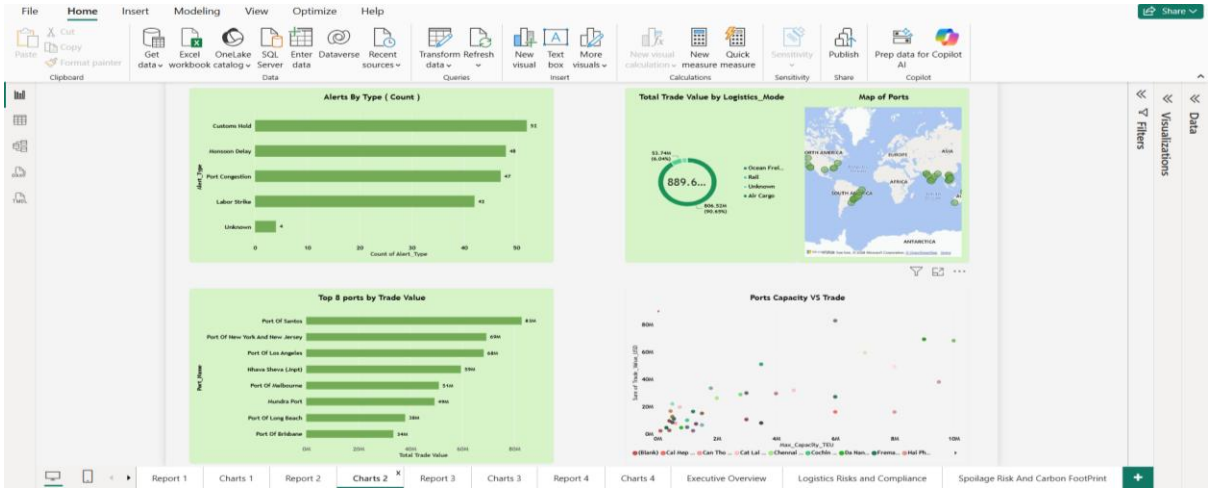
- Column Chart
- Bar Chart
- Donut
- Area Chart
- Map
- Scatter Chart
- Column And Line Chart
- Stacked Column Chart
- Pie chart
- Funnel Chart
- Waterfall Chart
- Ribbon Chart
- Tree Map

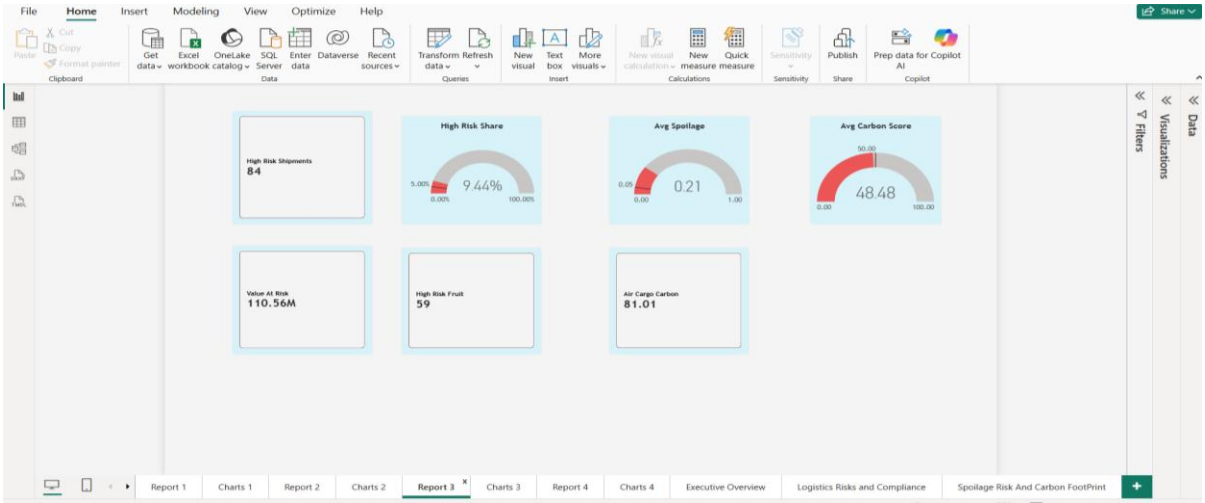
SCREENSHOTS OF REPORTS.

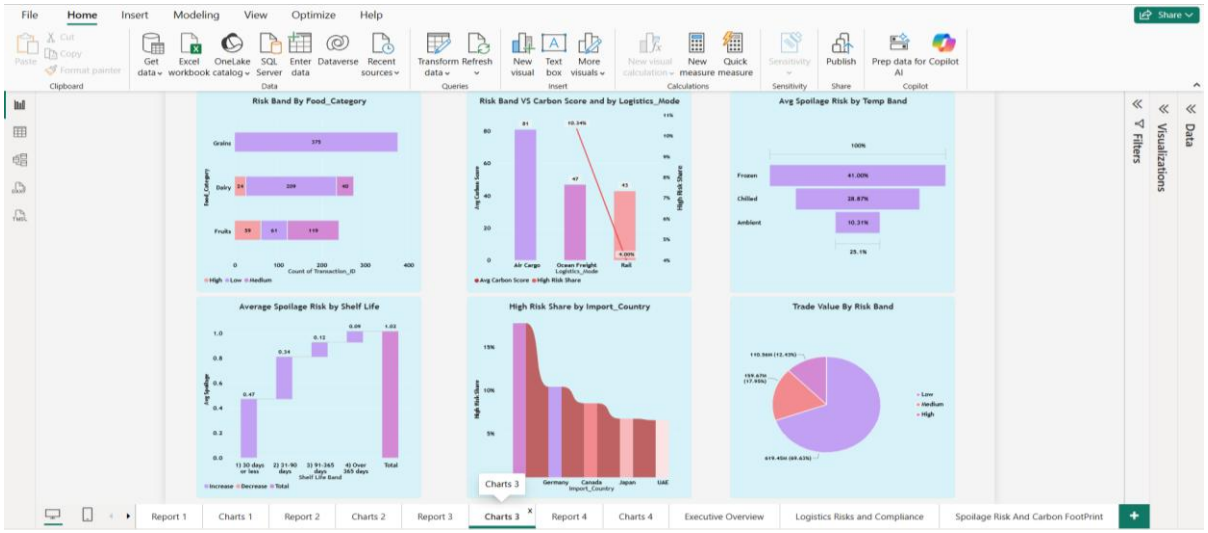


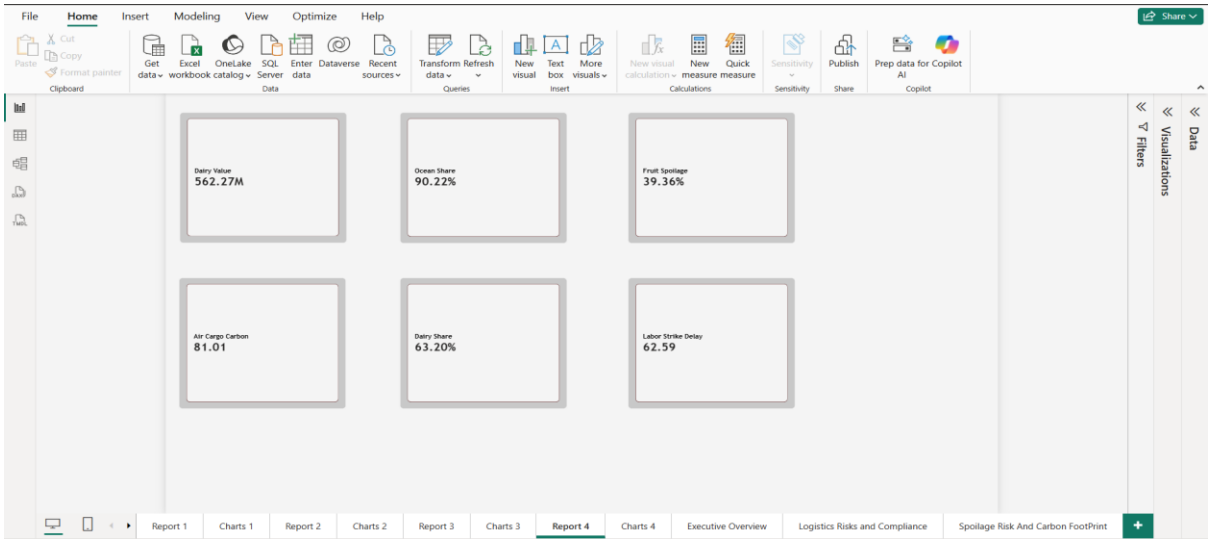


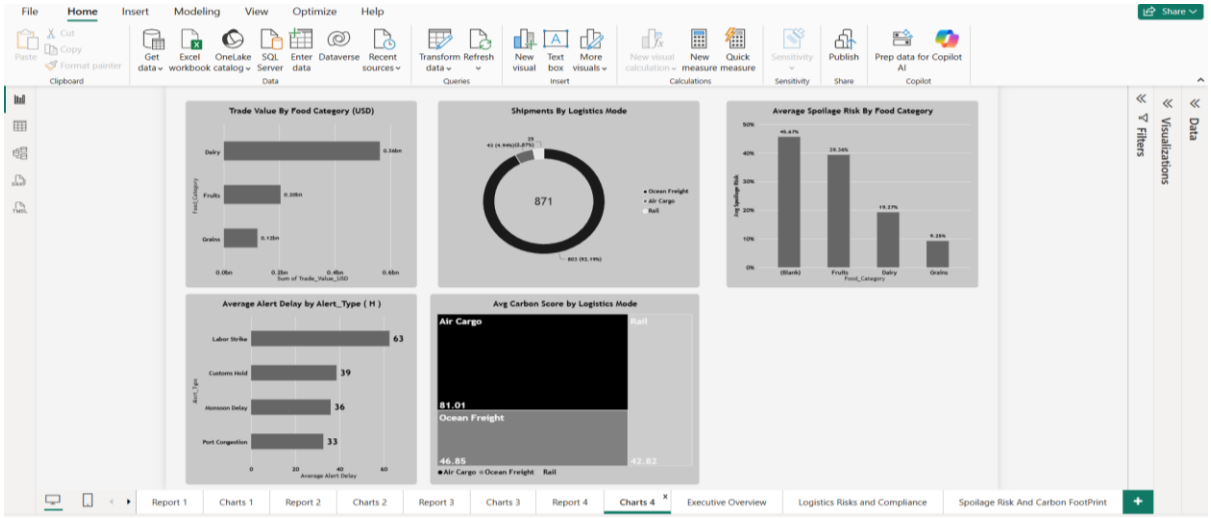

Logistics Alerts
193
Median Delay
38.25
Ports Used
38
Held/Quarantined
77
Average Alert Delay
24.00
41.89
0.00
72.00
Active Ports
28
Ocean Value Share
90.65%
Report 1
Charts 1
Report 2
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint


Alerts By Type (Count)
Customs Hold
Monsoon Delay
Port Congestion
Labor Strike
Unknown
Count of Alert_Type
Total Trade Value by Logistics_Mode
889.6...
Map of Ports
Top 8 ports by Trade Value
Port Of Santos
Port Of New York And New Jersey
Port Of Los Angeles
Port Of Melbourne
Hundra Port
Port Of Long Beach
Port Of Brisbane
Total Trade Value
Ports Capacity VS Trade
Report 1
Charts 1
Report 2
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint


High Risk Shipments
84
High Risk Share
9.44%
Avg Spoilage
0.21
Avg Carbon Score
48.48
Value At Risk
110.56M
High Risk Fruit
59
Air Cargo Carbon
81.01
Report 1
Charts 1
Report 2
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint


File
Home
Insert
Modeling
View
Optimize
Help
Share
Risk Band By Food_Category
Risk Band VS Carbon Score and by Logistics_Mode
Avg Spoilage Risk by Temp Band
Average Spoilage Risk by Shelf Life
High Risk Share by Import_Country
Trade Value By Risk Band
Charts 3
Report 1
Charts 1
Report 2
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Filters
Visualizations
Data

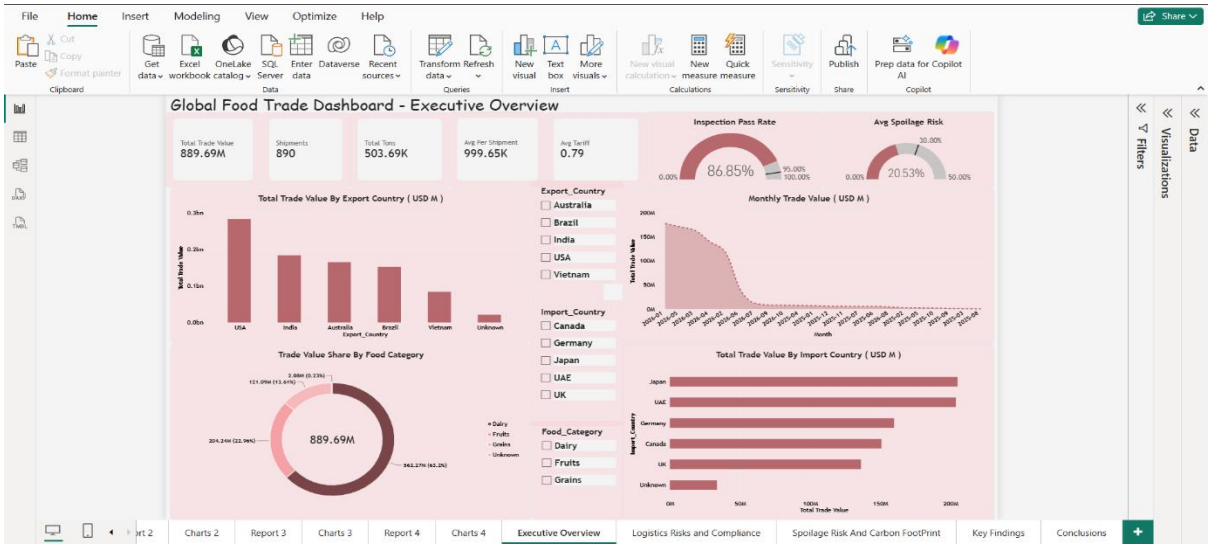

File
Home
Insert
Modeling
View
Optimize
Help
Share
Dairy Value
562.27M
Ocean Share
90.22%
Fruit Spoilage
39.36%
Air Cargo Carbon
81.01
Dairy Share
63.20%
Labor Strike Delay
62.59
Report 1
Charts 1
Report 2
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Filters
Visualizations
Data

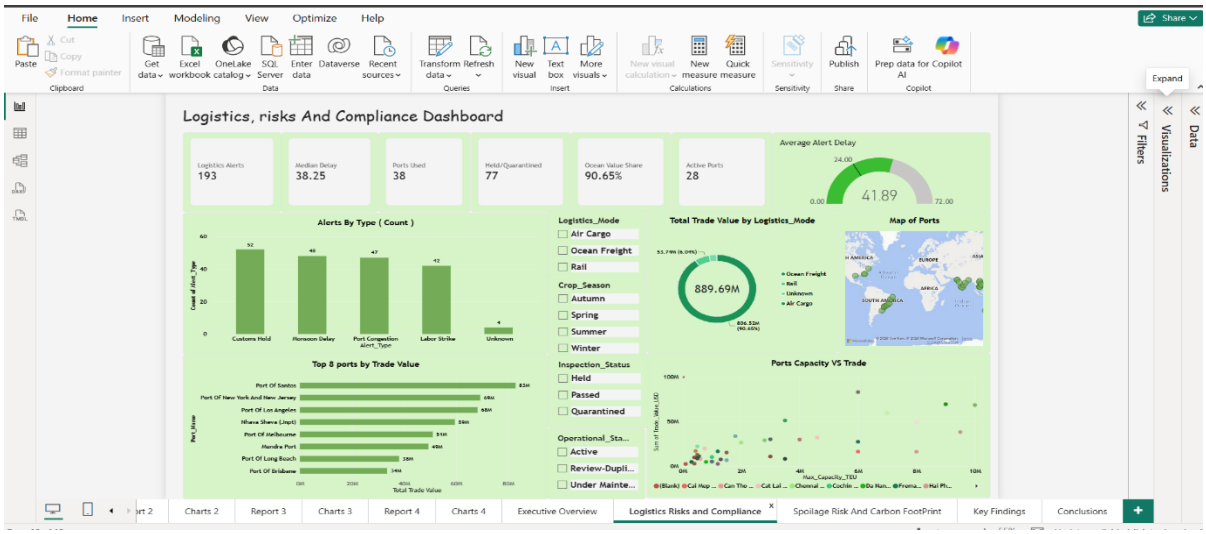

File
Home
Insert
Modeling
View
Optimize
Help
Share
Trade Value By Food Category (USD)
Shipments By Logistics Mode
871
Average Spoilage Risk By Food Category
Average Alert Delay by Alert_Type (H)
Labor Strike
63
Customs Hold
39
Monsoon Delay
36
Port Congestion
33
Avg Carbon Score by Logistics Mode
Air Cargo
81.01
Ocean Freight
46.85
42.82
Report 1
Charts 1
Report 2
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Filters
Visualizations
Data

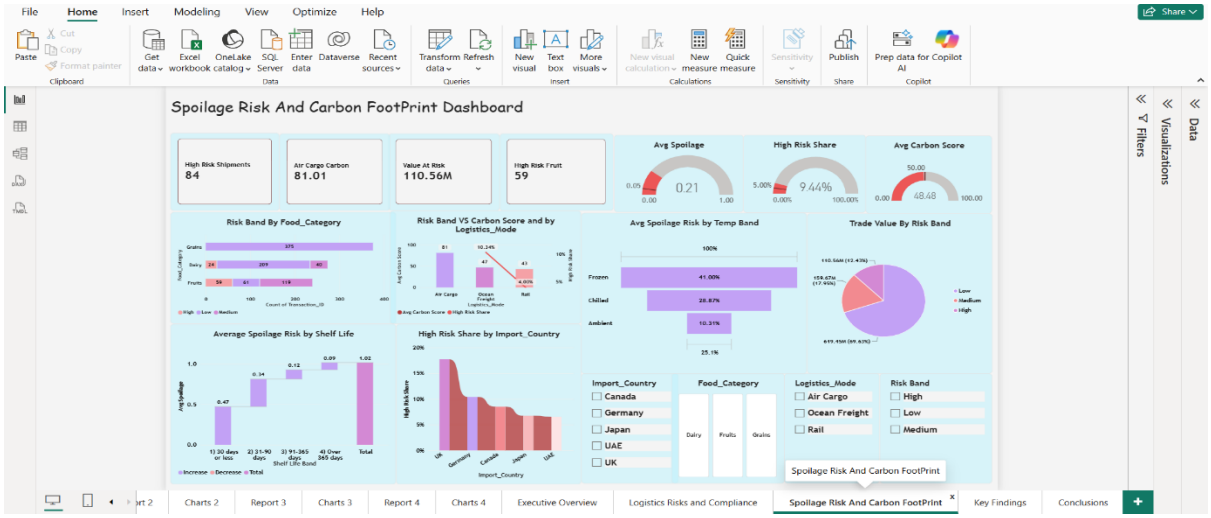
SCREENSHOTS OF DASHBOARDS.

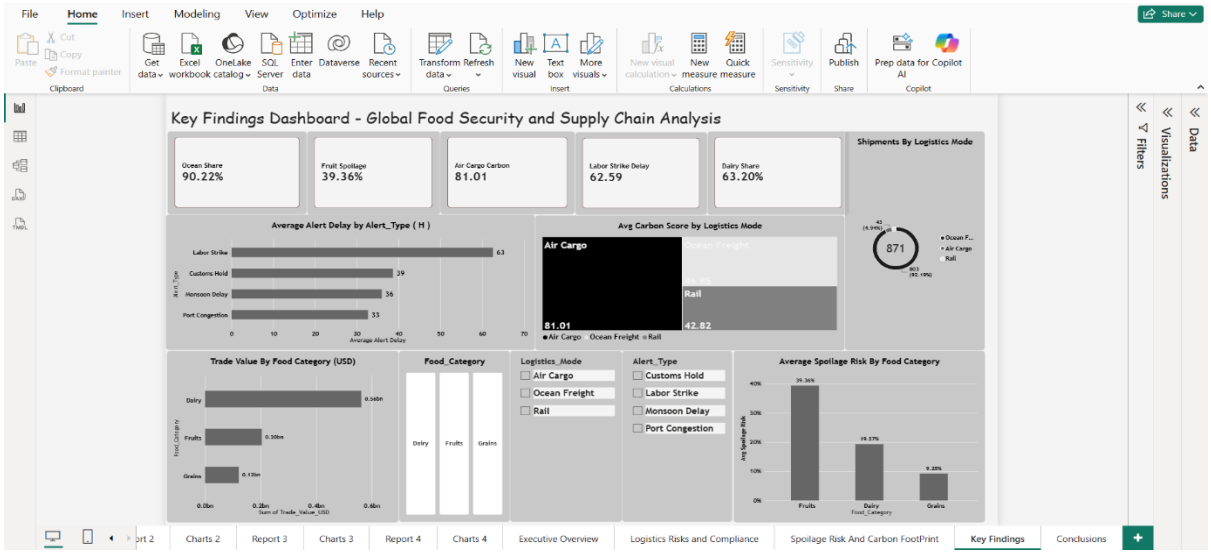
4 Dashboards are Created.

- Global Food Trade – Executive Overview
- Logistics , Risks and Compliance
- Spoilage Risk and Carbon Footprint
- Key Findings



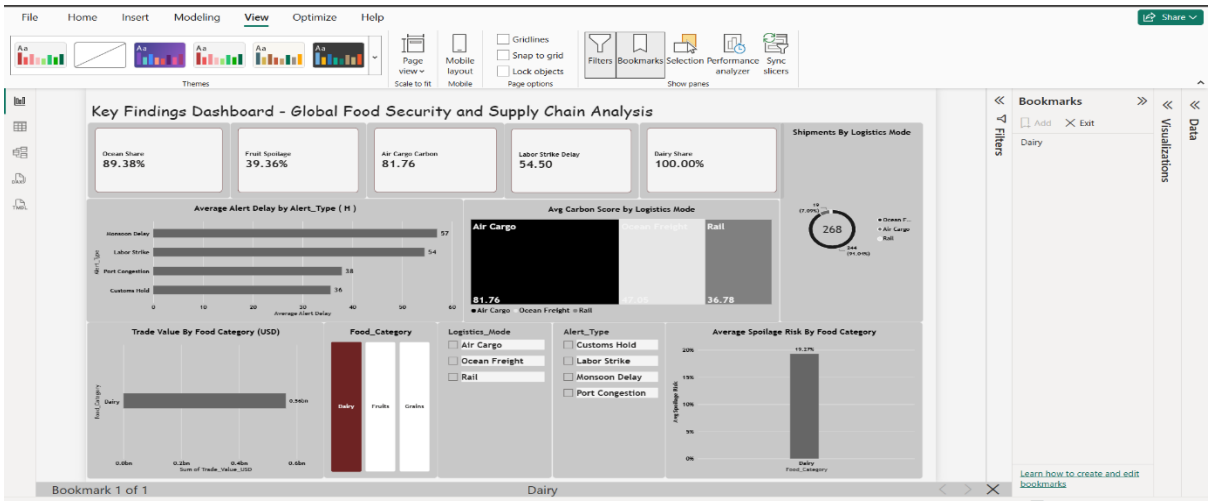


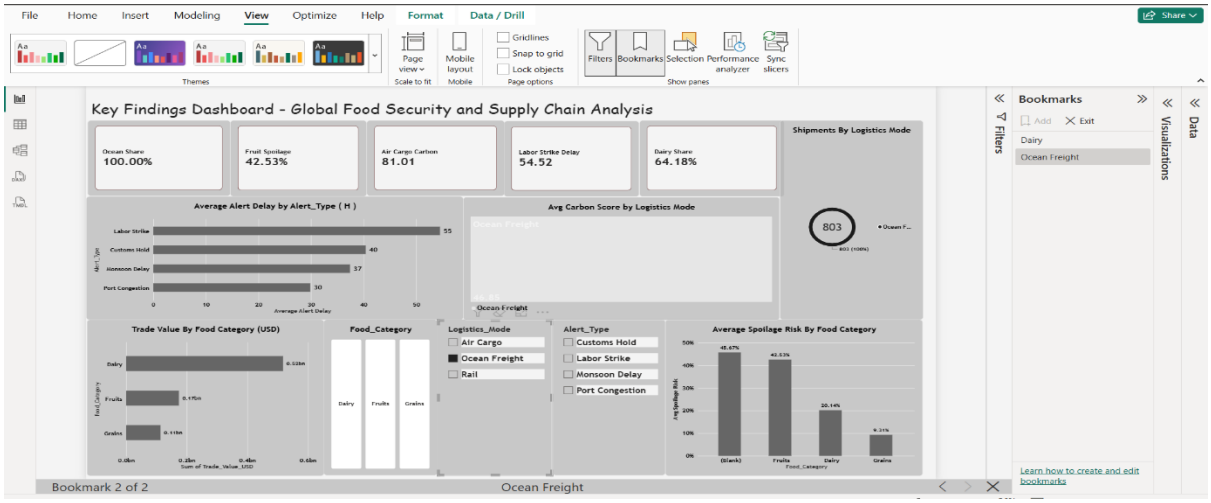

File
Home
Insert
Modeling
View
Optimize
Help
Share
Spoilage Risk And Carbon FootPrint Dashboard
High Risk Shipments
84
Air Cargo Carbon
81.01
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High Risk Fruit
59
Avg Spoilage
0.21
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48.48
Risk Band By Food_Category
Risk Band VS Carbon Score and by Logistics_Mode
Avg Spoilage Risk by Temp Band
Trade Value By Risk Band
Average Spoilage Risk by Shelf Life
High Risk Share by Import_Country
Import_Country
Canada
Germany
Japan
UAE
UK
Food_Category
Dairy
Fruits
Grains
Logistics_Mode
Air Cargo
Ocean Freight
Rail
Risk Band
High
Low
Medium
Spoilage Risk And Carbon FootPrint
Filters
Visualizations
Data
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Key Findings
Conclusions

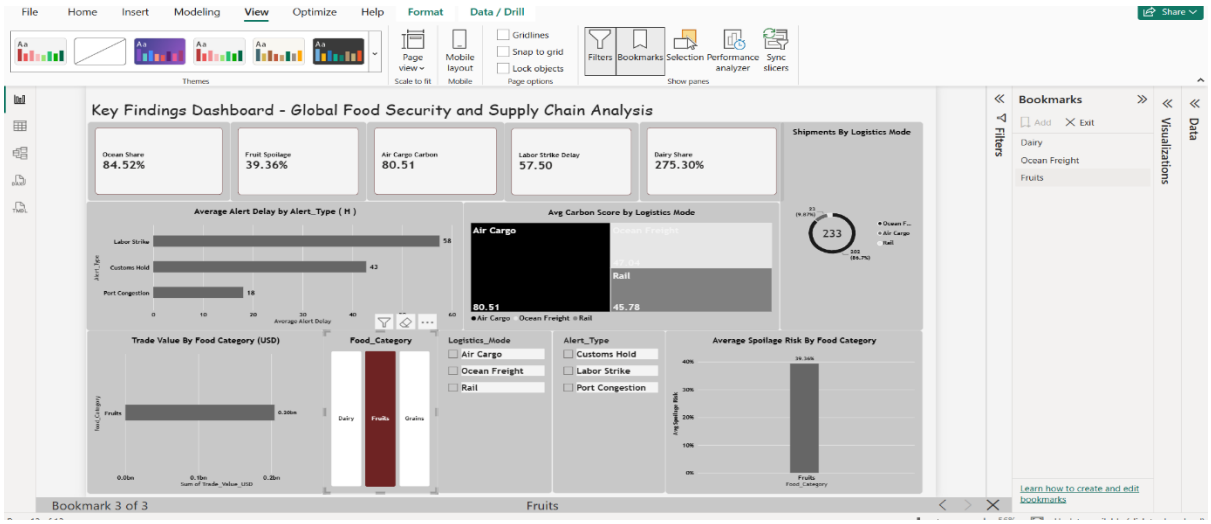

File
Home
Insert
Modeling
View
Optimize
Help
Share
Key Findings Dashboard - Global Food Security and Supply Chain Analysis
Ocean Share
90.22%
Fruit Spoilage
39.36%
Air Cargo Carbon
81.01
Labor Strike Delay
62.59
Dairy Share
63.20%
Shipments By Logistics Mode
871
Average Alert Delay by Alert_Type (H)
Labor Strike
Customs Hold
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Avg Carbon Score by Logistics Mode
Air Cargo
Rail
81.01
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Trade Value By Food Category (USD)
Dairy
Fruits
Grains
Food_Category
Logistics_Mode
Air Cargo
Ocean Freight
Rail
Alert_Type
Customs Hold
Labor Strike
Monsoon Delay
Port Congestion
Average Spoilage Risk By Food Category
Filters
Visualizations
Data
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Key Findings
Conclusions

BOOKMARKS.

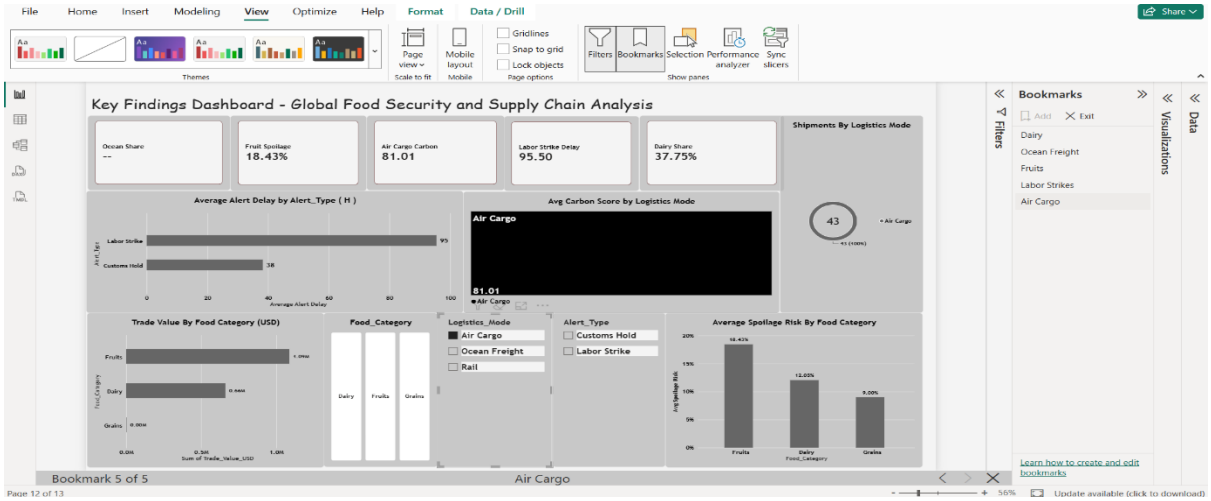
5 Bookmarks are Created to explain the Key Findings.











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9. Insights & Conclusions

Key Findings:

- Dairy has the highest trade value.
- Ocean Freight is the most used logistics mode.
- Fruits have the highest spoilage risk.
- Labor Strikes cause the highest average delays.
- Air Cargo has the highest carbon footprint.

Descriptive:

- 890 transactions were analyzed.
- Average trade value is about **$1 million**.
- USA is the top exporting country.

Diagnosis:

- Fruits face higher spoilage risk.
- Labor Strikes significantly increase shipment delays.
- Air Cargo has higher environmental impact.

Predictive:

- Perishable food shipments may face higher spoilage risk.
- Disrupted shipments may experience longer delays.
- High-value shipments need closer monitoring.

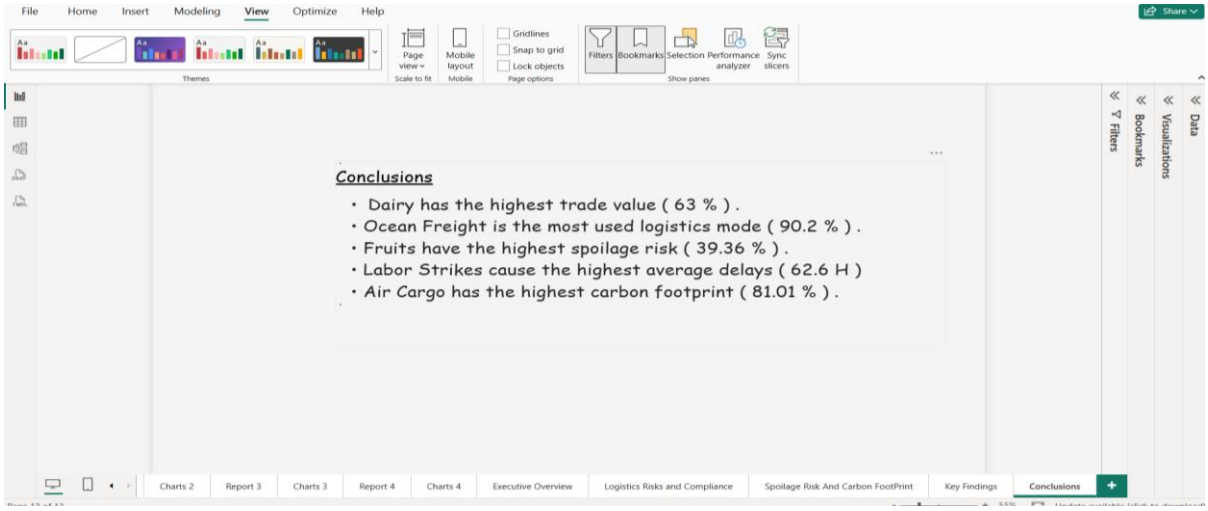
Prescriptive:

- Prioritize faster transport for perishable foods.
- Reduce dependency on Air Cargo where possible.
- Prepare contingency plans for Labor Strikes and other disruptions.
- Improve missing/unknown data values.

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10. Conclusions

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting. Cleaning the workbook in Excel left 890 valid transactions from 944 rows, and Power BI turned them into interactive dashboards with DAX measures, KPI cards, gauges And slicers .


File
Home
Insert
Modeling
View
Optimize
Help
Share
Themes
Page view
Scale to fit
Mobile layout
Mobile
Gridlines
Snap to grid
Lock objects
Page options
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Show panes
Conclusions
• Dairy has the highest trade value (63 %) .
• Ocean Freight is the most used logistics mode (90.2 %) .
• Fruits have the highest spoilage risk (39.36 %) .
• Labor Strikes cause the highest average delays (62.6 H)
• Air Cargo has the highest carbon footprint (81.01 %) .
Filters
Bookmarks
Visualizations
Data
Charts 2
Report 3
Charts 3
Report 4
Charts 4
Executive Overview
Logistics Risks and Compliance
Spoilage Risk And Carbon FootPrint
Key Findings
Conclusions